

JayaShankar Mangina

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SUMMARY

Software Engineer with 6+ years of experience designing scalable, cloud-native systems across Java, Spring Boot, and AWS backends, with additional depth in React and Angular frontends. Delivered 99%+ data accuracy, 75% latency reduction, and 100% test automation by building microservices, LLM-powered developer tools, and clean API layers. Passionate about system design, developer experience, and shipping high-performance products that scale reliably.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, TypeScript, JavaScript, React, Angular, Redux, RxJS, Spring Boot
Cloud & Infrastructure: AWS Lambda, AWS Step Functions, Docker, Git, Jenkins, CI/CD, Terraform, Kubernetes
Databases: MySQL, MongoDB, PostgreSQL, Amazon DynamoDB, Amazon S3
Testing & Quality: Jest, Cypress, JUnit, Test Automation, Integration Testing, Test-Driven Development (TDD)
Software Practices: Agile, System Design, RESTful API design, Microservices, Design Patterns

PROFESSIONAL EXPERIENCE

Software Engineer, Amazon – Austin, TX June 2024 - Present

- Developed a Step Functions workflow triggered by CLI to validate MCM-IDs and update site records, cutting entry errors by 99%.
- Eliminated schema-related bugs by implementing Pydantic attribute validation in API contracts, achieving 100% data integrity.
- Designed and implemented a recovery-aware deletion flow for stale site entries, improving latency by 75%.
- Engineered a microservice for automated network device discovery across Amazon warehouses, reducing manual IT troubleshooting by 90%.
- Automated on-call handover reports by integrating CTI data with an LLM tool, eliminating 100% of manual documentation.

Software Engineer, Amazon Web Services – Seattle, WA July 2022 - June 2024

- Engineered a carbon reporting engine that processed customer activity against public datasets, reducing emissions by 60% during internal pilot.
- Migrated DNS configurations to isolate production and non-production environments, eliminating 100% of cross-access risks with zero downtime.
- Streamlined JDK migration using LLMs to modernize two microservices, resulting in a 20% JVM performance boost and lower memory usage.
- Designed an integration test suite with 400+ JUnit tests simulating real customer scenarios, achieving 100% test coverage for emissions workflows.
- Implemented an end-to-end deletion feature for emissions data using Step Functions, Cloudscape UI, and Java backend logic, enabling real-time updates and seamless UI refresh.

Software Engineer, Cognizant Technology Solutions – Chennai, India Nov 2019 - Dec 2020

- Engineered a high-performance React dashboard with Redux, lazy loading, and code splitting, improving load time by 40% and enhancing user experience.
- Streamlined Angular rendering performance by 25% by fine-tuning ChangeDetectionStrategy and module loading.
- Implemented RxJS cancellation logic in Angular to stop redundant API calls during ticket switching, reducing bandwidth usage by 20%.
- Developed a micro-frontend infrastructure enabling isolated deployment pipelines with Webpack Module Federation, boosting dev productivity by 30%.

EDUCATION

Masters in Computer Science January 2021 - May 2022
Northwest Missouri State University – Maryville, MO GPA: 3.7/4

Bachelors in Electronics and Communication Engineering June 2015 - April 2019
Andhra University – Visakhapatnam, India GPA: 3.3/4

PROJECTS

CERT - Community Emergency Response Team (iOS)

<https://tinyurl.com/cert-ios>

- Developed a native iOS app in Swift for emergency responders, reducing incident processing time by 40% by integrating Firebase Firestore for real-time syncing and Firebase Auth for secure user sessions.
- Implemented a map-based incident reporting system using Core Location and MapKit, increasing report location accuracy by 60% through live GPS pinning and media attachments.
- Engineered backend data architecture using Firestore, Firebase Storage, and custom data schemas, improving report retrieval and scalability by 50%.
- Strengthened user access control via Firebase Authentication, delivering 70% improvement in user management and secure credential handling.

PicToSend – Peer-to-Peer Photo Share Application (iOS)

<https://tinyurl.com/pictosend-ios>

- Created an offline photo-sharing app in Swift using Multipeer Connectivity, enabling fast and secure peer-to-peer transfers without internet access.
- Reduced connection time by 50% by automating session discovery with MCBrowserViewController and MCAdvertiserAssistant, improving onboarding UX.
- Increased data transfer reliability by 60% through a fault-tolerant MCSession pipeline for sending images and text payloads.
- Built an intuitive UI with UICollectionView and lightweight state management, driving 40% higher engagement during photo sharing sessions.

PyFlink – Distributed PageRank Algorithm (Big Data)

<https://tinyurl.com/pyflink-bigdata>

- Designed and deployed a distributed PageRank algorithm using Java, Apache Beam, and Apache Flink, improving ranking precision by 20% through iterative convergence tuning.
- Reduced pipeline execution time by 30% by optimizing iteration counts (40→90) and parallelizing DoFn compute stages.
- Enabled fault-tolerance and scalability in stream processing using Apache Flink's windowing and checkpointing, ensuring robustness under distributed workloads.

WebSocketChat – Real-Time Chat Application (Web)

<https://tinyurl.com/websocketchat-web>

- Built a real-time chat platform with Spring Boot, WebSockets, and STOMP.js, enabling secure bi-directional messaging between users.
- Improved message delivery reliability by 70% via integration of a message broker and custom error-handling logic for dropped socket events.
- Developed a responsive UI in HTML/CSS/JS, increasing engagement by 40% with typing indicators, message threading, and clean chat flows.
- Strengthened data security by implementing session validation and event-based WebSocket listeners, reducing unauthorized access risk by 60%.

EchoMaster – Music Player Application (Android)

<https://tinyurl.com/echomaster-android>

- Built a full-featured Android music player in Kotlin, increasing playback reliability by 60% using MediaPlayer with robust error fallback for corrupted files.
- Created a synchronized audio visualizer using com.cleveroad:audiovisualization, raising session time and engagement by 50%.
- Implemented key playback controls (shuffle, repeat, seek) and playlist management features, driving a 40% increase in user retention metrics.

SpamText-Classifer – Naive Bayes Email Filter (ML)

<https://tinyurl.com/spamtextclassifier>

- Trained a Multinomial Naive Bayes classifier on 5,173 labeled emails (3,672 ham / 1,501 spam) using scikit-learn, achieving >95% test-set accuracy by vectorizing emails into 3,000-term bag-of-words and persisting the model.
- Engineered a dictionary-builder and email vectorizer in raw Python, processing all inputs in <30 seconds by streaming files and computing word frequencies without external NLP libraries.
- Delivered real-time CLI predictions in <100 ms by loading the pickled model in detector.py, vectorizing live input, and instantly returning ham/spam output.
- Built a from-scratch linear regression engine using NumPy, reducing mean squared error by 98% after 1,500 iterations, and visualized convergence using Matplotlib for interpretability.